



Work, Energy and Power · Mark Scheme

Further Mechanics 1 Chapter 2: Work, Energy and Power. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Work done and kinetic energy

- 1 $W = 20 \cos 30^\circ \times 5$ M1; = 86.6 J A1.
- 2 $\frac{1}{2}(4)(7^2) - \frac{1}{2}(4)(3^2)$ M1; = 80 J A1.
- 3 Work by force = $12 \times 10 = 120$ J B1; $F = 0.25 \times 2(9.8) = 4.9$ N M1; work against friction = 49 J A1.
- 4 Against gravity: $25(9.8)(8 \sin 20^\circ)$ M1; = 670 J A1; $F = 0.3 \times 25(9.8) \cos 20^\circ = 69.1$ N, work = 69.07×8 M1; = 553 J A1.
- 5(a) $W = 30 \cos 25^\circ \times 6$ M1; = 163 J A1.
- 5(b) $R = 3(9.8) - 30 \sin 25^\circ = 16.72$ N M1 A1; work = $0.4 \times 16.72 \times 6 = 40.1$ J A1.
- 5(c) $\frac{1}{2}(3)v^2 = 163.1 - 40.1$ M1 A1; $v = 9.06 \text{ ms}^{-1}$ A1.

Section B – Potential energy and work–energy

- 6 $mgh = 0.5(9.8)(12)$ M1; = 58.8 J A1.
- 7 $\frac{1}{2}m(14^2) = mgh$, $h = 10$ m M1 A1; $\frac{1}{2}v^2 = \frac{1}{2}(196) - 9.8(5)$, $v = 9.90 \text{ ms}^{-1}$ A1.
- 8 $\frac{1}{2}v^2 = \frac{1}{2}(3^2) + 9.8(2.5)$ M1; $v = 7.62 \text{ ms}^{-1}$ A1; the acceleration is not constant (the path is curved, so the direction and size of the resultant force change) B1.
- 9 Loss in GPE = $2(9.8)(5 \sin 30^\circ) = 49$ J M1; work against friction = $0.2 \times 2(9.8) \cos 30^\circ \times 5 = 16.97$ J M1; $\frac{1}{2}(2)v^2 = 49 - 16.97$ M1; $v = 5.66 \text{ ms}^{-1}$ A1.
- 10(a) $\sin \alpha = 0.6$, $\cos \alpha = 0.8$ B1; $R = 4(9.8)(0.8) = 31.36$, $F = \frac{1}{4}R = 7.84$ N M1; $\frac{1}{2}(4)(10^2) = 4(9.8)(0.6)d + 7.84d$ M1 A1; $200 = 31.36d$, $d = 6.38$ m A1.
- 10(b) Component of weight down the plane = 23.52 N, $F_{\max} = 7.84$ N M1; $23.52 > 7.84$, so P does not remain at rest (it slides back down) A1.
- 10(c) $\frac{1}{2}(4)v^2 = 23.52d - 7.84d$ M1 A1; $= 15.68 \times 6.378 = 100$, so $v = \sqrt{50} = 7.07 \text{ ms}^{-1}$ A1.

Section C – Power

- 11 $F = \frac{P}{v} = \frac{12000}{20}$ M1; = 600 N A1.
- 12 $F = \frac{24000}{15} = 1600$ N B1; $1600 - 800 = 1000a$ M1; $a = 0.8 \text{ ms}^{-2}$ A1.
- 13 At maximum speed $F = 600$ N, $P = 600 \times 40 = 24$ kW B1; $\frac{24000}{20} - 600 = 900a$ M1; $a = 0.667 \text{ ms}^{-2}$ A1.
- 14 Weight component = $5000(9.8) \left(\frac{1}{20}\right) = 2450$ N M1; driving force $F = \frac{60000}{v}$ M1; at maximum speed $\frac{60000}{v} = 2000 + 2450$ M1; $v = 13.5 \text{ ms}^{-1}$ A1.
- 15(a) $\frac{36000}{40} = R$ M1; $R = 900$ A1.
- 15(b) $\frac{36000}{v} = 900 + 1200(9.8) \left(\frac{1}{14}\right)$ M1 A1; = 1740, so $v = 20.7 \text{ ms}^{-1}$ A1.
- 15(c) $F = \frac{36000}{12} = 3000$ N B1; $3000 - 1740 = 1200a$ M1; $a = 1.05 \text{ ms}^{-2}$ A1.



Roar-level challenge

16 $\frac{18000}{30} = 30k \text{ Mt}; k = 20 \text{ At}; \text{ up the hill } \frac{18000}{v} = 20v + 800(9.8) \left(\frac{1}{16}\right) = 20v + 490 \text{ Mt}; 20v^2 + 490v - 18000 = 0 \text{ At};$
 $v = \frac{-490 + \sqrt{1680100}}{40} = 20.2 \text{ ms}^{-1} \text{ At}.$